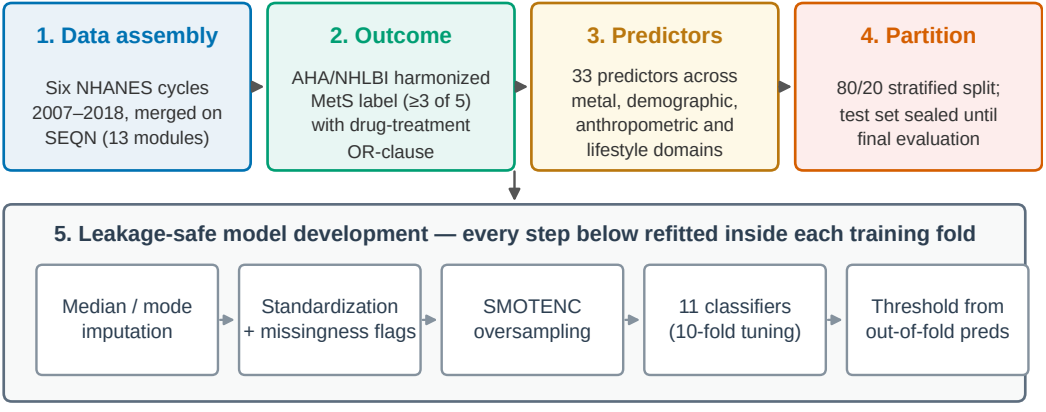
1. Data assembly
Six NHANES cycles 2007–2018, merged on SEQN (13 modules)
2. Outcome
AHA/NHLBI harmonized MetS label (≥3 of 5) with drug-treatment OR-clause
3. Predictors
33 predictors across metal, demographic, anthropometric and lifestyle domains
4. Partition
80/20 stratified split; test set sealed until final evaluation
5. Leakage-safe model development — every step below refitted inside each training fold
Median / mode imputation
Standardization + missingness flags
SMOTENC oversampling
11 classifiers (10-fold tuning)
Threshold from out-of-fold preds